and the parameter

$$w(r) := \sup_{C_\varphi(r)\times C_\psi(r)} W + w_{V,U}(r) + w^{U,V}(r) - \nu_V(V).$$

Note that for any $r \geqslant r_0$ we have

$$\begin{aligned}\nu_V\left(C_\psi(r)\right) &= e^{-V_\star}\ \nu\left(e^{-(V-V_\star)}\ 1_{(V-V_\star)\leqslant \delta^{-1}\log r}\right)\\ &\geqslant w_0(r) := e^{-V_\star}\ e^{-(\delta^{-1}\log r)}\ \nu\left(C_\psi(r)\right) > 0\end{aligned}$$

and this yields, for any $x \in C_\varphi(r)$ and $r \geqslant r_0$, the estimate

$$\mathcal{S}_{2n}(x,dy) \geqslant \varepsilon_0(r)\ \mathcal{Y}_r(dy) \quad \text{with} \quad \varepsilon_0(r) := w_0(r)\ e^{-w(r)}.$$

In the same vein, for any $y \in C_\psi(r)$ we have

$$\mathcal{S}_{2n+1}(y,dx) \quad \geqslant \quad e^{\nu_V(V)}\ e^{-(W(x,y)+\mathcal{W}_{V,U}(x)+\mathcal{W}^{U,V}(y))}\ \lambda_U(dx)\ 1_{C_\varphi(r)}(x)$$

and, arguing as above, we arrive at

$$\lambda_U\left(C_\varphi(r)\right) \geqslant w_1(r) := e^{-U_\star}\ e^{-(\delta^{-1}\log r)}\ \lambda\left(C_\varphi(r)\right) > 0.$$

Thus, for any $y \in C_\psi(r)$ we have

$$\mathcal{S}_{2n+1}(y,dx) \geqslant \varepsilon_1(r)\ \mathcal{X}_r(dx)$$

where

$$\varepsilon_1(r) := w_1(r)\ e^{-w(r)} \quad \text{and} \quad \mathcal{X}_r(dx) := \frac{\lambda_U(dx)\ 1_{C_\varphi(r)}(x)}{\lambda_U\left(C_\varphi(r)\right)}.$$

The above estimates ensure that the local minorization condition (4.14) is met with $n_0 = 1$ and the function

$$\varepsilon(r) := \varepsilon_0(r) \wedge \varepsilon_1(r) > 0.$$

Hence, we conclude that Condition $\mathcal{D}_{\varphi,\psi}(\mathcal{S})$ is satisfied with $n_0 = 1$.

Using (2.8) and (D.3), for any $n \geqslant p \geqslant 1$ we verify that

$$\frac{d\pi_{2n}}{d\nu_V}(y) - 1 = \left[(\delta_y - \nu_V)\left(\mathcal{S}^\circ_{2n}\ldots\mathcal{S}^\circ_{2(p+1)}\right)\right]\left(\frac{d\pi_{2p}}{d\nu_V}\right),$$

which implies the inequality

$$\left|\frac{d\pi_{2n}}{d\nu_V}(y) - 1\right| \leqslant \|\frac{d\pi_{2p}}{d\nu_V}\|_\psi\ \left\|\!\left|(\delta_y - \nu_V)\left(\mathcal{S}^\circ_{2n}\ldots\mathcal{S}^\circ_{2(p+1)}\right)\right|\!\right\|_\psi.$$

On the other hand, combining Proposition D.7 with the norm equivalence (3.4) and Theorem 4.6, we see that there exists some $\kappa > 0$ and some constants $c_{\kappa,i} > 0$, $i = 1,2$, such that

$$\begin{aligned}\left\|\!\left|(\delta_y - \nu_V)\left(\mathcal{S}^\circ_{2n}\ldots\mathcal{S}^\circ_{2(p+1)}\right)\right|\!\right\|_\psi &\leqslant c_{\kappa,1}\ \left\|\!\left|(\delta_y - \nu_V)\left(\mathcal{S}^\circ_{2n}\ldots\mathcal{S}^\circ_{2(p+1)}\right)\right|\!\right\|_{\psi_\kappa}\\ &\leqslant c_{\kappa,1}\ \rho^{2(n-p)}\ |\delta_y - \nu_V|_{\psi_\kappa} \leqslant c_{\kappa,2}\ \rho^{2(n-p)}\ (\psi(y) + \nu_V(\psi)).\end{aligned}$$

Therefore, for any $n \geqslant 1$, we obtain

$$\left\|\frac{d\pi_{2n}}{d\nu_V} - 1\right\|_\psi \leqslant c_{\kappa,3}\ \rho^{2n} \quad \text{and, similarly,} \quad \left\|\frac{d\nu_V}{d\pi_{2n}} - 1\right\|_\psi \leqslant c_{\kappa,4}\ \rho^{2n}, \tag{D.16}$$

for some constants $c_{\kappa,i}$, $i = 3,4$. By the same argument, using (2.8) we have, for any $n \geqslant p \geqslant 1$,

$$\frac{d\pi_{2n+1}}{d\lambda_U}(x) - 1 = \left[(\delta_x - \lambda_U)(\mathcal{S}^\circ_{2n+1}\ldots\mathcal{S}^\circ_{2p+1})\right]\left(\frac{d\pi_{2p-1}}{d\lambda_U}\right)$$

and arrive at

$$\left\| \frac{d\pi_{2n+1}}{d\lambda_U} - 1 \right\|_\varphi \vee \left\| \frac{d\lambda_U}{d\pi_{2n+1}} - 1 \right\|_\varphi \leqslant c_{\kappa,5}\ \rho^{2n} \tag{D.17}$$

for any $n \geqslant 1$ and some constant $c_{\kappa,5}$.

Finally, note that for any function $f(x) > 0$ we have

$$\begin{aligned} |\log f(x)| &= \log\left(1 + (f(x) - 1)\right)\ 1_{f(x)\geqslant 1} + \log\left(1 + \frac{1}{f(x)} - 1\right)\ 1_{1/f(x)\geqslant 1} \\ &\leqslant (f(x) - 1)\ 1_{f(x)\geqslant 1} + \left(\frac{1}{f(x)} - 1\right)\ 1_{1/f(x)\geqslant 1}, \end{aligned}$$

where the last assertion comes from the fact that $\log(1 + u) \leqslant u$, for any $1 + u > 0$. This implies that

$$\|\log f\|_\varphi = \|\log 1/f\|_\varphi \leqslant \|f - 1\|_\varphi + \|1/f - 1\|_\varphi.$$

The inequality above combined with (D.16) and (D.17) yields the estimates

$$\left\| \log \frac{d\pi_{2n}}{d\nu_V} \right\|_\psi \vee \left\| \log \frac{d\pi_{2n+1}}{d\lambda_U} \right\|_\varphi \leqslant c_{6,\kappa}\ \rho^{2n},$$

for some finite constant $c_{\kappa,6}$. This completes the proof of (4.24). ∎

Appendix E. Some technical proofs

Proof of (D.1) and (D.2). We first note, using (2.2) and (2.3), that

$$\frac{d\mathcal{P}_{2n}}{d\mathcal{P}_{2n+1}}(x, y) = \frac{d\pi_{2n}}{d\nu_V}(y) = \exp\left(V_{2n+1}(y) - V_{2n}(y)\right) \tag{E.1}$$

and, in the same vein,

$$\frac{d\mathcal{P}_{2n+1}}{d\mathcal{P}_{2(n+1)}}(x, y) = \frac{d\pi_{2n+1}}{d\lambda_U}(x) = \exp\left(U_{2(n+1)}(x) - U_{2n+1}(x)\right). \tag{E.2}$$

If we additionally observe that

$$\begin{aligned} \mathcal{P}_{2(n+1)}(d(x, y)) &= \frac{d\lambda_U}{d\pi_{2n+1}}(x)\ \mathcal{P}_{2n}(d(x, y))\ \frac{d\nu_V}{d\pi_{2n}}(y) \quad \text{and} \\ \mathcal{P}_{2n+1}(d(x, y)) &= \frac{d\lambda_U}{d\pi_{2n-1}}(x)\ \mathcal{P}_{2n-1}(d(x, y))\ \frac{d\nu_V}{d\pi_{2n}}(y) \end{aligned}$$

then, using (2.1), we obtain the formulae

$$\begin{aligned} \mathcal{S}_{2(n+1)}(x, dy) &= \frac{d\lambda_U}{d\pi_{2n+1}}(x)\ \mathcal{S}_{2n}(x, dy)\ \frac{d\nu_V}{d\pi_{2n}}(y) \quad \text{and} \\ \mathcal{S}_{2n+1}(y, dx) &= \frac{d\nu_V}{d\pi_{2n}}(y)\ \mathcal{S}_{2n-1}(y, dx)\ \frac{d\lambda_U}{d\pi_{2n-1}}(x), \end{aligned}$$

and the commutation properties now follow from elementary manipulations. ∎

Proof of (4.10). Assume Condition $\mathcal{D}_{\varphi,\psi}(\mathcal{S}^\circ)$ is satisfied. Combining (4.6) with (2.8) we arrive at

$$\nu_V(\psi) = \nu_V(\mathcal{S}^\circ_{2n}(\psi)) \leqslant \epsilon\ \nu_V(\psi) + c \quad \text{which implies} \quad \nu_V(\psi) \leqslant \frac{c}{1-\epsilon}.$$

In a similar way, one can also show that

$$\lambda_U(\varphi) \leqslant \frac{c}{1-\epsilon}.$$

Using the Markov evolution equations in (2.9) together with (4.6) we find that, for any $n \geqslant n_0$,

$$\begin{aligned}\pi_{2n}(\psi) &= \pi_{2(n-1)}(\mathcal{S}^\circ_{2n}(\psi)) \\ &\leqslant \epsilon\ \pi_{2(n-1)}(\psi) + c \leqslant \epsilon^{n-n_0}\ \pi_{2n_0}(\psi) + \frac{c}{1-\epsilon}\end{aligned}$$

and, by the same argument,

$$\begin{aligned}\pi_{2n+1}(\varphi) &= \pi_{2n-1}(\mathcal{S}^\circ_{2n+1}(\varphi)) \\ &\leqslant \epsilon\ \pi_{2n-1}(\varphi) + c \leqslant \epsilon^{n-n_0}\ \pi_{2n_0+1}(\varphi) + \frac{c}{1-\epsilon}.\end{aligned}$$

■

Proof of Proposition 5.3. It is straightforward to show that

$$\begin{aligned}2W_U(y) &= \left\|\tau^{-1/2}(y-m_0)\right\|_F^2 + \mathrm{Tr}(\tau^{-1}\sigma_\beta) + \log\det(2\pi\tau) \quad \text{and} \\ 2W^V(x) &= \left\|\tau^{-1/2}(\beta(x-m) + (m_0-\overline{m}))\right\|_F^2 + \ \mathrm{Tr}(\tau^{-1}\overline{\sigma}) + \log\det(2\pi\tau),\end{aligned}$$

with $m_0 := (\alpha + \beta m)$ and $\sigma_\beta := (\beta\sigma\beta')$, hence W_U and W^V are locally bounded. Moreover, for any $\delta \geqslant 0$ we obtain

$$\begin{aligned}2(\delta U - W^V) &= -\mathrm{Tr}(\tau^{-1}\overline{\sigma}) + \delta\ \log\det(2\pi\sigma) - \log\det(2\pi\tau) \\ &\quad +(x-m)'\left(\delta\ \sigma^{-1} - \beta'\tau^{-1}\beta\right)(x-m) \\ &\quad -2(m_0-\overline{m})'\tau^{-1}\beta(x-m) - (m_0-\overline{m})'\tau^{-1}(m_0-\overline{m})\end{aligned}$$

and, in the same way,

$$\begin{aligned}2(\delta V - W_U) &= -\mathrm{Tr}(\tau^{-1}\sigma_\beta) + \delta\ \log\det(2\pi\overline{\sigma}) - \log\det(2\pi\tau) \\ &\quad +(y-\overline{m})'\left(\delta\ \overline{\sigma}^{-1} - \tau^{-1}\right)(y-\overline{m}) \\ &\quad -2(\overline{m}-m_0)'\tau^{-1}(y-\overline{m}) - (\overline{m}-m_0)'\tau^{-1}(\overline{m}-m_0).\end{aligned}$$

Therefore,

$$\mathcal{V}_\delta = \delta V - W_U \in \mathcal{B}_\infty(\mathbb{R}^d) \quad \text{and} \quad \mathcal{U}_\delta = \delta U - W^V \in \mathcal{B}_\infty(\mathbb{R}^d)$$

if, and only if,

$$\beta'\tau^{-1}\beta < \delta\ \sigma^{-1} \quad \text{and} \quad \tau^{-1} < \delta\ \overline{\sigma}^{-1}.$$

On the other hand, straightforward manipulations yield

$$\begin{aligned}2(W^V - (1-\delta)U) &= \mathrm{Tr}(\tau^{-1}\overline{\sigma}) - (1-\delta)\ \log\det(2\pi\sigma) + \log\det(2\pi\tau) \\ &\quad -(x-m)'\left((1-\delta)\ \sigma^{-1} - \beta'\tau^{-1}\beta\right)(x-m) \\ &\quad +2(m_0-\overline{m})'\tau^{-1}\beta(x-m) + (m_0-\overline{m})'\tau^{-1}(m_0-\overline{m}),\end{aligned}$$

which shows that

$$\lambda_{(1-\delta)U}(e^{W^V}) < \infty \Longleftrightarrow \beta'\tau^{-1}\beta < (1-\delta)\ \sigma^{-1}.$$